From Unit Distances and Sum–Product to Polynomial-Image Compression

AI-Assisted Mathematical Research

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Abstract

How often can one distance occur among finitely many points in the plane? Erdős's unit-distance problem asks for the largest possible number of pairs exactly one unit apart. An arithmetic construction by OpenAI produces more such pairs than the near-linear conjecture allowed. A related construction for the sum–product problem produces finite real sets with relatively few sums and products. Both arguments arrange many exact repetitions. The first needs enough pairs realizing one distance. The second must control every value in two images.

This report reconstructs the two constructions and uses them to study polynomial images in the setting of Elekes and Rónyai. The main theorem concerns each fixed polynomial $P(x, y) = (ax + by)^2 + cx + ey$ with integer coefficients. It constructs finite real sets A of arbitrarily large cardinality for which $|P(A, A)| \leq |A|^{2-\varepsilon_P}$ with a fixed positive exponent saving. An auxiliary polynomial restricts the outputs to square residues at selected primes. Counting algebraic integers uniformly in each allowed class bounds the entire image, and a common scaling returns to the fixed polynomial. The family includes examples outside both exceptional forms of Elekes–Rónyai. Clearing denominators and normalizing the output extends the bound to every fixed rational-coefficient quadratic whose homogeneous quadratic part has rank one.

AI assisted the reading of the source proofs, the proposal of constructions and the checking of symbolic identities and exact finite examples. In the extensions, a transfer theorem identifies sufficient hypotheses for a full-rank quadratic, and a unit-population theorem supplies many candidate symmetries. Neither yet yields the required complete-image estimate. Cubic examples show how one large fiber can coexist with an almost quadratic image, while a quartic estimate shows where the cost of bounding outputs consumes a local saving. These distinctions shaped how we used AI. Its proposals became constructions when we could prove both the growth of the inputs and a bound on every output.

1 Background, research objective and methodology

1.1 The unit-distance problem

The unit-distance problem asks how many pairs among n distinct planar points can be exactly one unit apart. Introduced by Erdős in 1946, it measures repetitions of one prescribed distance, whereas the distinct-distances problem counts the different distances determined by a point set [18]. For a finite set $V \subset \mathbb{R}^2$, define

$$\nu(V) = \#\big\{\{p, q\} \subseteq V : p \neq q,\ \|p - q\|_2 = 1\big\}.$$

Here # denotes cardinality and $\|p - q\|_2$ is Euclidean distance. Each unordered pair is counted once. The extremal function

$$\nu(n) = \max_{\substack{V \subset \mathbb{R}^2 \\ |V| = n}} \nu(V)$$

is the largest number of unit-distance pairs possible among n planar points. Exhibiting one configuration gives a lower bound for this maximum. An upper bound must hold for every configuration.

Erdős's lattice construction gives

$$\nu(n) \geq n^{1+c/\log\log n}$$

for some absolute $c > 0$ and all sufficiently large n [18, Theorem 2]. All logarithms are natural. The notation $o(1)$ denotes a quantity tending to zero as $n \to \infty$. Since $c/\log\log n \to 0$, this is compatible with his near-linear conjecture $\nu(n) = n^{1+o(1)}$, which predicts that for every fixed $\varepsilon > 0$, eventually $\nu(n) \leq n^{1+\varepsilon}$. The stronger formulation considered in the OpenAI manuscript is

$$\nu(n) \leq n^{1+C/\log\log n}$$

for some fixed $C > 0$ and all sufficiently large n [1]. It would match the lattice construction up to constants in the exponent. It specifies a rate of approach to exponent one that the near-linear conjecture does not, so the two statements are not equivalent.

In 1984, Spencer, Szemerédi and Trotter proved the universal bound $\nu(n) = O(n^{4/3})$, meaning that $\nu(n)$ is at most a fixed constant times $n^{4/3}$ for all sufficiently large n [19]. This remained far above the lattice lower bound and did not reach either conjectured upper bound. The original OpenAI prompt later requested a *proof or disproof* of the stronger formulation. It asked whether absolute C, N work for every $n \geq N$, or whether each choice of C, N admits a counterexample [1, p. 3].

OpenAI's construction disproves both the near-linear unit-distance conjecture $\nu(n) = n^{1+o(1)}$ and its stronger quantitative formulation $\nu(n) \leq n^{1+C/\log\log n}$. The construction supplies a fixed $\delta > 0$ and finite planar sets V_j, with $n_j = |V_j| \to \infty$, satisfying

$$\boxed{\nu(n_j) \geq \nu(V_j) \geq n_j^{1+\delta}.}$$

Choosing $\varepsilon = \delta/2$ contradicts the near-linear conjecture. For every fixed C, eventually $C/\log\log n_j < \delta$, contradicting the stronger formulation [1, Theorem 1.1].

The contradiction only requires an unbounded sequence of sizes n_j. What matters for the project is how the examples are built. Arithmetic supplies directions of Euclidean length one, and geometry realizes many of them as differences in one finite set. The sum–product construction uses related tools to control the complete sets of values generated by addition and multiplication.

1.2 Sum–product, Elekes–Rónyai and polynomial-image compression

For a finite set $A \subset \mathbb{R}$, its sumset and product set are

$$A + A = \{a + b : a, b \in A\}, \qquad AA = \{ab : a, b \in A\}.$$

An arithmetic progression such as $\{1, \ldots, n\}$ has only $2n - 1$ sums, while a geometric progression $\{1, r, \ldots, r^{n-1}\}$, with $r > 1$, has only $2n - 1$ products. These examples show how additive or multiplicative structure can make many input pairs produce the same value. The sum–product question asks whether both kinds of repetition can occur strongly on the same set.

A near-quadratic sum–product conjecture over the reals predicted that, for every $\varepsilon > 0$,

$$\max\{|A + A|, |AA|\} \geq c_\varepsilon |A|^{2-\varepsilon},$$

for every finite $A \subset \mathbb{R}$, where $c_\varepsilon > 0$ depends on ε, not on A. Bloom, Sawin, Schildkraut and Zhelezov disproved this by constructing arbitrarily large finite real sets for which

$$\max\{|A + A|, |AA|\} \leq |A|^{2-c} \tag{1}$$

with a fixed $c > 0$ [3]. Their construction combines additive and multiplicative structure in number fields. Its use of real algebraic numbers does not resolve the corresponding conjecture for integer sets.

Addition and multiplication are the simplest examples of bivariate polynomial maps. More generally, for a polynomial P and finite sets $A, B \subset \mathbb{R}$, define

$$P(A, B) = \{P(a, b) : a \in A,\ b \in B\}.$$

We write $A^2 = A \times A$ for the set of all ordered pairs, including (a, a). Thus $P(A, A) = \{P(a, b) : (a, b) \in A^2\}$. The product set AA and the set of individual squares $\{a^2 : a \in A\}$ are sets of numbers. The *fiber* over an output z consists of the input pairs $(a, b) \in A \times B$ with $P(a, b) = z$.

Our project seeks particular input sets on which a given polynomial has substantially fewer values than the trivial upper bound $|A|^2$. A power saving means a fixed decrease in the exponent two. A fixed factor such as $1/2$ multiplying $|A|^2$ gives no such decrease. The objective is

$$\boxed{\begin{array}{c}\text{For a fixed polynomial } P, \text{ find } \varepsilon_P > 0 \\ \text{and finite sets } A_j \subset \mathbb{R}, \\ |A_j| \longrightarrow \infty, \qquad |P(A_j, A_j)| \leq |A_j|^{2-\varepsilon_P}.\end{array}} \tag{2}$$

The polynomial and the positive exponent remain fixed as the sets grow. The same set must appear in both inputs, and the bound must account for every ordered pair.

The Elekes–Rónyai problem asks how the algebraic form of P determines the size of this image. The exceptional forms in its expansion theory are

$$P(x, y) = g(u(x) + v(y)) \qquad \text{and} \qquad P(x, y) = g(u(x)v(y)),$$

where each of g, u, v is a polynomial in one variable. For instance, $(x + y)^2$ has the additive form with $u(t) = v(t) = t$ and $g(t) = t^2$. These expressions retain an underlying additive or multiplicative structure. Outside these forms, the expansion theorems give lower bounds forcing $P(A, B)$ to grow faster than linearly on every sufficiently large Cartesian product of equal-sized sets [4]. This

universal lower bound can coexist with the upper bound in (2). An image can grow faster than linearly and still have a fixed saving below quadratic growth.

The family studied in this report is

$$P(x, y) = (ax + by)^2 + cx + ey,$$

where a, b, c, e are arbitrary fixed integers. Pohoata's construction for $(x - y)^2 + x + y$ and Liu's for $(x - y)^2 + Mx$, with a fixed integer M, use the same square-leading, split-prime and lattice-counting mechanism [20, Theorem 1.1] [21, Theorem 1.2 and §5]. These papers supply the literature context for the family studied here.

Section 3 proves a power-saving bound for every fixed integer-coefficient polynomial of the displayed form. The proof combines sparse output residues, uniform lattice counting and the same scaling in both inputs. The normalization argument in Section 3.6 extends the conclusion to every fixed rational-coefficient quadratic whose homogeneous quadratic part has rank one. Within the displayed integer family, the additional condition $ab(ae - bc) \neq 0$ identifies a nonspecial subfamily. Section 4 examines full-rank quadratic and higher-degree problems that remain unresolved in this project.

1.3 AI-assisted methodology and attribution

AI helped interpret difficult proofs, formulate candidates, work through symbolic identities and organize exact finite calculations [5, 6, 7, 8]. Its role was to help turn promising ideas into statements that could be checked.

C0 is the project record for the quadratic construction. Its search was conducted offline using the materials supplied at that stage. Later branches also included targeted source consultation, as documented in the norm–trace research record [11, §4]. Literature comparisons made while preparing this report are distinguished from the original derivation. The clarification of coefficient hypotheses in Section 3 is also distinct from the original investigation: the historical C0 theorem was stated under stronger assumptions.

The source authors describe different roles for AI. OpenAI's manuscript reports an internally generated solution followed by AI assessment and human mathematical checking and expository revision [1, pp. 2–3]. The companion *Remarks* is a separate human-authored mathematical treatment [2]. The sum–product authors describe GPT-5.5 Pro as an early sounding board, credit almost all main ideas to human work, and identify an AI suggestion used in their unit-separation lemma [3, The role of AI in this proof]. These attributions describe how the source papers were produced.

The archived quadratic-search prompt requires: "The same set must occur in both inputs" [7, §1]. It also asks for an upper bound on all outputs and an account of the factors that weaken it. The proposal introduced an auxiliary polynomial whose residues were easier to control, then used a common scaling to recover the fixed polynomial. The C0 correction record distinguished totally real input fields from the complex fields of the unit-distance proof, and replaced an unsupported *exact* output radius with the proved upper bound [10]. The subsequent prompt explicitly warns that a failed estimate is not an impossibility theorem [8, §1]. These prompt, proposal and correction stages are documented.

AI suggestions were candidate arguments, not evidence of correctness. Symbolic calculations clarified identities; exact finite examples exposed weaknesses in estimates. In the extensions, a proved implication specifies what a candidate set would achieve if it met the stated hypotheses. Finding sets that meet those hypotheses is a separate construction problem. Exact counts for finitely many sets can test a proposal while leaving that problem open.

1.4 Arithmetic tools for the constructions

A finite set of real algebraic numbers can be difficult to count on the real line. The constructions count it in several coordinates first, then use one embedding to realize the same set on the line or in the plane. Consider

$$F = \mathbb{Q}(\sqrt{2}) = \{m + n\sqrt{2} : m, n \in \mathbb{Q}\}.$$

Its degree is two because $1, \sqrt{2}$ form a basis over $\mathbb{Q}$. Its two embeddings into $\mathbb{R}$ send $m + n\sqrt{2}$ to $m + n\sqrt{2}$ and $m - n\sqrt{2}$. Each preserves sums, products and the number one. If it sent a nonzero element to zero, multiplication by the image of its inverse would give $1 = 0$. Thus one coordinate already determines the element.

An algebraic integer satisfies a monic polynomial with integer coefficients. In this field the ring of integers is $O_F = \mathbb{Z}[\sqrt{2}]$. In the two coordinates together its image is a lattice,

$$\Phi_F(m + n\sqrt{2}) = m(1, 1) + n(\sqrt{2}, -\sqrt{2}), \qquad m, n \in \mathbb{Z}.$$

The displayed vectors span a parallelogram of area $2\sqrt{2}$, the absolute value of their determinant. This area is the *covolume*, the volume of one fundamental cell. The discriminant is its square, $\Delta_F = 8$. The two coordinates are conjugates of the same algebraic integer. In fact, in just one coordinate $\mathbb{Z}[\sqrt{2}]$ has nonzero elements $(\sqrt{2} - 1)^m$ tending to zero. Retaining both coordinates is what gives a discrete lattice.

For a totally real field F of degree d, write its real embeddings as $\sigma_{F,1}, \ldots, \sigma_{F,d}$. Their combined map is

$$\Phi_F : F \longrightarrow \mathbb{R}^d, \qquad \alpha \longmapsto (\sigma_{F,1}(\alpha), \ldots, \sigma_{F,d}(\alpha)).$$

The covolume of $\Phi_F(O_F)$ is $\sqrt{\Delta_F}$. If the root discriminant $\mathrm{rd}(F) = \Delta_F^{1/d}$ is at most D, this volume is at most $D^{d/2}$. Dividing the volume of a box by the covolume gives its *average* lattice-point count over translations. The point count in a particular box also depends on its position.

The field norm multiplies the conjugates. In the example,

$$N_{F/\mathbb{Q}}(m + n\sqrt{2}) = (m + n\sqrt{2})(m - n\sqrt{2}) = m^2 - 2n^2.$$

In every number field a nonzero algebraic integer has nonzero integral norm. Applied to a difference, this implies that distinct algebraic integers in a totally real field differ by at least one in absolute value at some embedding. This separation allows small cubes centered on their vectors to be counted by volume.

A unit is an algebraic integer with an integral inverse. Its absolute norm is one. The map $\mathrm{Log}_F(u) = (\log|\sigma_{F,k}(u)|)_{k=1}^d$ sends products to sums in the hyperplane $\sum_k t_k = 0$. For instance $1 + \sqrt{2}$ has inverse $\sqrt{2} - 1$, and its log vector is $(h, -h)$ with $h = \log(1 + \sqrt{2})$. By Dirichlet's unit theorem, these vectors form a lattice of rank $d - 1$, and the two units differing by sign have the same vector. The conventional regulator $\mathrm{Reg}(F)$ gives Euclidean hyperplane covolume $\sqrt{d}\ \mathrm{Reg}(F)$ [3, §3.2]. A logarithmic box therefore counts multiplicative choices. The sum–product proof below uses quantitative estimates for that count.

We use F_j for a totally real field in the construction under discussion and $d_j = [F_j : \mathbb{Q}]$. Each construction specifies its own fields. In the planar problem $K_j = F_j(i)$ supplies complex directions. In the polynomial construction a single real embedding $\iota_j : F_j \to \mathbb{R}$ places the chosen inputs on the line. We use all embeddings to count and separate algebraic integers, then one injective embedding to obtain the real or planar set required by the problem. Appendices A and B record the coordinate and volume conventions used in each setting.

2 Source constructions: unit distances and sum–product

To obtain many unit distances, it suffices to realize enough prescribed differences among the points of one planar set. To make a sumset and a product set small, one must control the values produced by *all* pairs of a real set. Both constructions exploit a field of growing degree. Their different counting requirements determine which parts of each argument can be used for polynomial images.

2.1 A planar construction with many unit distances

Construction and arithmetic input. We seek a fixed $\delta > 0$ and finite planar sets V_j with $n_j = |V_j| \to \infty$ and $\nu(V_j) \geq n_j^{1+\delta}$. OpenAI's arithmetic input supplies totally real fields F_j of degrees $d_j \to \infty$ and extensions $K_j = F_j(i)$. A fixed set of primes splits completely in all the F_j. Its product Q and a constant $\gamma > 0$ give finite collections of directions satisfying

$$U_j \subset Q^{-2}\mathcal{O}_{K_j}, \qquad |U_j| \geq e^{\gamma d_j}, \qquad u\,c(u) = 1 \quad (u \in U_j),$$

where c fixes F_j and sends i to $-i$. These fields, primes and constants are supplied together [1, Propositions 2.2 and 3.8]. The directions can have denominators. For instance, $(2+i)/(2-i) = (3+4i)/5$ in $\mathbb{Q}(i)$ gives the planar unit vector $(3/5, 4/5)$. This illustrates a single direction.

At one layer, write F, K, d, U for F_j, K_j, d_j, U_j. Choose one complex embedding $\sigma_{K,k}$ above each real embedding of F, one from each conjugate pair. The map $\Phi_K(\beta) = (\sigma_{K,1}(\beta), \ldots, \sigma_{K,d}(\beta))$ embeds K in $\mathbb{C}^d$. Let $\Lambda = \Phi_K(Q^{-2}\mathcal{O}_K)$ be its fractional-ideal lattice and $W_r = \{z \in \mathbb{C}^d : |z_k| \leq r \text{ for all } k\}$ the product of d closed discs of radius r. The first-coordinate projection is $\pi_1 \colon \mathbb{C}^d \to \mathbb{C}$, $\pi_1(z) = z_1$. The candidate sets are

$$X_{\mathbf{a}} = (\mathbf{a} + \Lambda) \cap W_r, \qquad V_{\mathbf{a}} = \pi_1(X_{\mathbf{a}}) \subset \mathbb{C} \cong \mathbb{R}^2. \tag{3}$$

They are finite since the lattice is discrete and the window is bounded. We will fix r and choose a successful translate $\mathbf{a}$ in each layer.

Verification and quantitative bounds. Projection onto one coordinate can identify different points of $\mathbb{C}^d$ when $d > 1$. On this translated lattice it retains every point. Indeed, $\pi_1 \circ \Phi_K = \sigma_{K,1}$, so the image of $\mathbf{a} + \Phi_K(\beta)$ is the complex number $a_1 + \sigma_{K,1}(\beta)$, viewed through its real and imaginary parts. If two such images agree, the translate cancels and $\sigma_{K,1}(\beta - \beta') = 0$. Injectivity of the field embedding gives $\beta = \beta'$. Thus $|V_{\mathbf{a}}| = N_{\mathbf{a}} := |X_{\mathbf{a}}|$, even though $\mathbf{a}$ can be any vector in $\mathbb{C}^d$.

Projection also gives the desired length for the selected differences. Each embedding sends $c(u)$ to the complex conjugate of $\sigma_{K,k}(u)$, since F is totally real. Hence $|\sigma_{K,k}(u)|^2 = \sigma_{K,k}(uc(u)) = 1$. Let $E_{\mathbf{a}}$ count ordered pairs $(x, y) \in X_{\mathbf{a}} \times X_{\mathbf{a}}$ with $y - x = \Phi_K(u)$ for some $u \in U$. Each projects to distinct endpoints a unit distance apart, and injectivity distinguishes the ordered pairs. There are at most two orientations for each unordered pair, so $\nu(V_{\mathbf{a}}) \geq E_{\mathbf{a}}/2$. This length calculation uses one complex coordinate. The length of the original vector $\Phi_K(u)$ in $\mathbb{C}^d$ is $\sqrt{d}$.

To select the translate, average $\mathbf{a}$ uniformly over a half-open fundamental cell T of Λ. The translates $T + \lambda$, $\lambda \in \Lambda$, tile the ambient real space. For any bounded measurable region S, integrating $\sum_{\lambda \in \Lambda} \mathbf{1}_S(\mathbf{a} + \lambda)$ over T thus gives $\mathrm{vol}(S)$. Dividing by $\mathrm{vol}(T) = \mathrm{covol}(\Lambda)$ makes the integral an average. In particular,

$$\mathbb{E}N_{\mathbf{a}} = \frac{\mathrm{vol}(W_r)}{\mathrm{covol}(\Lambda)}.$$

For a fixed u, the endpoint $x + \Phi_K(u)$ belongs to the same translated lattice as x. Both points lie in the window exactly when the starting point belongs to $W_r \cap (W_r - \Phi_K(u))$. Distinct directions give different differences. Summing the same integral over u gives

$$\mathbb{E}E_{\mathbf{a}} = \frac{\sum_{u\in U} \operatorname{vol}\big(W_r \cap (W_r - \Phi_K(u))\big)}{\operatorname{covol}(\Lambda)}.$$

Let ρ_r be the area of overlap of two radius-r discs with centers one unit apart, divided by πr^2. Each coordinate displacement has modulus one. Multiplication of the overlap areas therefore gives $\operatorname{vol}(W_r \cap (W_r - \Phi_K(u))) = \rho_r^d \operatorname{vol}(W_r)$. In particular, $\mathbb{E}E_{\mathbf{a}} = |U|\rho_r^d \mathbb{E}N_{\mathbf{a}}$. Some nonempty translate has $E_{\mathbf{a}} \geq |U|\rho_r^d N_{\mathbf{a}}$. Otherwise integrating over nonempty translates would give a strict reverse inequality, since $\mathbb{E}N_{\mathbf{a}} > 0$ and empty translates contribute zero to both counts. The comparison uses the two expectations directly and requires no average of the ratio $E_{\mathbf{a}}/N_{\mathbf{a}}$. The overlap fraction tends to one as r grows. Fix $r > 1$ large enough that $\log \rho_r > -\gamma/2$. At every layer a selected translate then satisfies $E_{\mathbf{a}} \geq e^{\gamma d/2} N_{\mathbf{a}}$ [1, Lemma 2.4].

To turn this gain per degree into a gain per point, count the points before projecting. A difference of distinct lattice points has the form $\Phi_K(Q^{-2}\beta)$ with $0 \neq \beta \in \mathcal{O}_K$. Pairing conjugate embeddings gives

$$\prod_{k=1}^{d} |\sigma_{K,k}(Q^{-2}\beta)| = Q^{-2d}|N_{K/\mathbb{Q}}(\beta)|^{1/2} \geq Q^{-2d}.$$

At least one coordinate has modulus at least Q^{-2}. Thus open product discs of radius $Q^{-2}/2$ about the points of $X_{\mathbf{a}}$ are pairwise disjoint and lie inside $W_{r+Q^{-2}/2}$. Comparing volumes in $2d$ real dimensions yields

$$N_{\mathbf{a}} \leq (1 + 2rQ^2)^{2d} = e^{Bd}, \qquad B = 2\log(1 + 2rQ^2).$$

The separating coordinate can depend on the pair. Only the ambient count uses this separation [1, Lemmas 2.5–2.6].

In each layer take $V_j = V_{\mathbf{a}_j}$ from (3) for a translate satisfying the edge estimate. Since $E_{\mathbf{a}_j} \leq n_j^2$ and $n_j = N_{\mathbf{a}_j} > 0$, we have $n_j \geq e^{\gamma d_j/2}$. The sizes therefore grow, and the upper count gives $d_j \geq \log n_j/B$. Consequently

$$\nu(V_j) \geq \tfrac{1}{2} n_j e^{\gamma d_j/2} \geq \tfrac{1}{2} n_j^{1+\gamma/(2B)} \geq n_j^{1+\gamma/(4B)}$$

for all sufficiently large layers. The last step absorbs the factor $1/2$ into the smaller fixed gain.

The arithmetic behind the directions now has a concrete role. From t fixed split primes the source obtains 2^{td} choices of ideals. Selecting ideals in one class loses a factor of at most the class number $h(K) \leq H^d$. The tower supplies $t\log 2 - \log H = \gamma > 0$ before d grows. Appendix B explains how the surviving choices yield distinct directions. The boundary loss above is small enough to turn this supply into actual edges in one finite planar set.

2.2 A real construction with small sumsets and product sets

Construction and arithmetic input. Bloom, Sawin, Schildkraut and Zhelezov construct finite $A \subset \mathbb{R}$ of arbitrarily large size n with both $|A + A|$ and $|AA|$ at most n^{2-c} for a fixed $c > 0$ [3, Theorem 1.1]. Their field input supplies totally real fields F of arbitrarily large degree $d \geq 2$, with discriminant $\Delta_F \leq D^d$ for a fixed D. This construction uses bounded root discriminant without requiring particular primes to split [3, Theorem 3.2].

In such a field let $B_F^+(T)$ consist of the algebraic integers p with $|\sigma_{F,k}(p)| \le T$ at every real embedding. Let $B_F^\times(T)$ consist of the integral units g with $|\log|\sigma_{F,k}(g)|| \le T$ at every embedding. Choose $X, Y \ge 2$ independently of the field degree, with X integral. We specify their required sizes after the image estimates. Fix a sufficiently small absolute $\eta > 0$ as in the proof of [3, Lemma 4.1], with $\eta < (\varphi - 1)/(\varphi + 1)$ for $\varphi = (1 + \sqrt{5})/2$, and take $\eta X \ge 1$. Choose a real embedding $\iota_F \colon F \hookrightarrow \mathbb{R}$. The construction is

$$\begin{aligned} \mathcal{P} &= X + B_F^+(\eta X), \qquad G = B_F^\times(Y), \\ \mathcal{A} &= G\mathcal{P} = \{gp : g \in G,\ p \in \mathcal{P}\}, \qquad A = \iota_F(\mathcal{A}) \subset \mathbb{R}. \end{aligned} \tag{4}$$

The superscript + denotes the additive box. Translating it by X puts each conjugate of $p \in \mathcal{P}$ in the positive interval $[(1-\eta)X, (1+\eta)X]$. The set G supplies units with multiplicative structure but limited distortion at each embedding. Since ι_F is an injective field map, it preserves cardinalities of $\mathcal{A}$ and of its two complete images.

Verification and quantitative bounds. First we check that different representations gp are distinct. If a unit u satisfies $\varphi^{-1} < |\sigma_{F,k}(u)| < \varphi$ for all k, then all conjugates of the algebraic integer $(u - u^{-1})^2$ lie in $[0, 1)$. Its norm is an integer of absolute value less than one. It must vanish, giving $u^2 = 1$ and $u = \pm 1$ [3, Lemma 3.4]. A collision $g_1 p_1 = g_2 p_2$ gives $g_1/g_2 = p_2/p_1$. Every conjugate of the right side lies between $(1-\eta)/(1+\eta)$ and its reciprocal, strictly inside the separation interval. Positivity rules out -1, so $g_1 = g_2$ and $p_1 = p_2$. We may now use the actual input size $n = |A| = |G||\mathcal{P}|$.

Products benefit from the logarithmic description of G. Every product of two units in G lies in $B_F^\times(2Y)$, and $|\mathcal{P}\mathcal{P}| \le |\mathcal{P}|^2$. Thus

$$GG \subseteq B_F^\times(2Y), \qquad \frac{|AA|}{n^2} \le \frac{|GG|\,|\mathcal{P}|^2}{|G|^2|\mathcal{P}|^2} = \frac{|GG|}{|G|^2}.$$

The count of units in the two logarithmic boxes will bound this ratio. For sums, each conjugate of gp has modulus at most $(1+\eta)Xe^Y < 2Xe^Y$. The norm separation of algebraic integers in all real coordinates gives

$$\mathcal{A} + \mathcal{A} \subseteq B_F^+(4Xe^Y), \qquad |A + A| \le (8Xe^Y + 1)^d.$$

Enlarging Y helps multiplication, but it enlarges this additive radius by e^Y.

The quantitative counts in [3, Lemma 4.1] combine the additive lattice estimate with the unit count and $\mathrm{Reg}(F) \le \Delta_F$. Appendix B.3 gives the two lattice estimates. With the fixed choice of η above, one absolute $c_0 > 0$ gives

$$\begin{aligned} \frac{(c_0 XY)^d}{Y\Delta_F^{3/2}} \le n \le (XY/c_0)^d, \\ |AA| \le c_0^{-d} Y^{1-d} \Delta_F^2 n^2, \\ |A + A| \le (e^Y/c_0)^d \Delta_F^{1/2} n. \end{aligned}$$

The second image bound is linear in n, with an exponential coefficient. To compare it with n^2, divide by n^2 and use the lower estimate for n once. Substituting $\Delta_F \le D^d$ throughout gives

$$\begin{aligned} \frac{|AA|}{n^2} \le Y\left(\frac{D^2}{c_0 Y}\right)^d, \qquad \frac{|A+A|}{n^2} \le Y\left(\frac{D^2 e^Y}{c_0^2 XY}\right)^d, \\ n \ge \frac{1}{Y}\left(\frac{c_0 XY}{D^{3/2}}\right)^d, \qquad n \le (XY/c_0)^d. \end{aligned} \tag{5}$$

Fix D from the field supply. Choose a fixed Y large enough for the lemma and for $D^2/(c_0Y) < 1$. Next choose the integer X large enough for the lemma, for $D^2e^Y/(c_0^2XY) < 1$, and for $c_0XY/D^{3/2} > 1$. The last inequality ensures $n \to \infty$ as the degree grows. Increasing X also ensures $B = \log(XY/c_0) > 0$. If $q < 1$ is the larger of the two image bases in (5), fix $0 < \gamma < -\log q$. For all sufficiently large degrees $Yq^d \le e^{-\gamma d}$. Since $n \le e^{Bd}$, both complete images satisfy

$$\max\{|A + A|, |AA|\} \le n^2 e^{-\gamma d} \le n^{2-\gamma/B}.$$

This exponent applies to the same real set for both operations. Its proof needed uniqueness of the inputs and a count of every output for each operation.

2.3 From the source constructions to a polynomial-image strategy

For a fixed polynomial P, our objective (2) concerns the distinct values on the whole square $A^2 = A \times A$. Selected unit-distance pairs can motivate a search for repetitions, while the sum–product proof suggests counting complete images. A bound for a new polynomial must be established on its own inputs.

For $z \in P(A, A)$, write $r_P(z) = |\{(a, b) \in A^2 : P(a, b) = z\}|$. Grouping ordered pairs according to their output gives

$$|P(A, A)| = \sum_{(a,b)\in A^2} \frac{1}{r_P(P(a, b))}. \tag{6}$$

A fiber of size r contributes r terms of size $1/r$. To use this identity as an upper bound, one must also account for pairs in small fibers.

For example, take $P(x, y) = (x - y)^2$. On $A = \{1, \dots, n\}$ its values are precisely k^2 for $0 \le k < n$, so there are n of them. On $A = \{2^i : 0 \le i < n\}$ every positive difference $2^j - 2^i = 2^i(2^{j-i} - 1)$, $j > i$, determines its indices. Its largest power-of-two divisor gives i, and the odd factor gives $j - i$. Squaring these distinct positive integers preserves their distinctness. Including zero, the image has $1 + n(n - 1)/2$ values. The same zero fiber of size n occurs in both examples, even though the second image is nearly quadratic. Appendix B.4 gives the related inequality for the total collision count.

Another way to bound the entire image is to enclose every auxiliary output in a finite set whose size we can bound [6]. Suppose all outputs belong to at most M_d residue classes and satisfy bounds on their conjugates. If each allowed class contains at most K_d possible values satisfying those bounds, their total number is at most M_dK_d. The uniform upper bound in each class makes any assumption of equidistribution unnecessary. The actual finite inputs may reach only some of the allowed residues. The count still covers every pair.

The reference quantity is the number of actual input pairs, $|A|^2$. A parametrization with repeated descriptions contributes only its distinct elements. If restriction reduces each scalar input count by a factor $e^{\eta d}$, the pair count falls by $e^{2\eta d}$. A common affine change with nonzero slope preserves the cardinality and the Cartesian square. Unequal shifts or a map mixing the two coordinates require a separate argument to recover one input set.

Quantitatively, suppose $n_d = |A_d| \to \infty$ and that, after these costs have been included, fixed $B, \gamma > 0$ give $n_d \le e^{Bd}$ and $|P(A_d, A_d)|/n_d^2 \le e^{-\gamma d}$. From $d \ge (\log n_d)/B$ it follows that

$$\boxed{|P(A_d, A_d)| \le n_d^{2-\gamma/B}.} \tag{7}$$

If the logarithmic bounds include $o(d)$ errors, the bound holds eventually for every fixed saving exponent $0 < \varepsilon < \gamma/B$, since the positive gap $\gamma - \varepsilon B$ absorbs those errors per degree. A square of a linear form suggests a way to meet this comparison. Its residues occupy at most about half the classes at each large odd prime, and its quadratic growth matches the scale of the pair count. Section 3 makes both estimates work for one fixed polynomial and one growing sequence of real sets.

3 A power-saving construction for the square-plus-linear family

The C0 report developed a construction for a family of nonspecial quadratics [10, Theorem C0.1 and §§ 4–6]. Its counting argument gives the following broader formulation.

3.1 The theorem and the real input sets

Theorem 1 (Compression for a square of a linear form plus a linear term)**.** *For every fixed $a, b, c, e \in \mathbb{Z}$, let*

$$\ell_1(x, y) = ax + by, \qquad \ell_2(x, y) = cx + ey, \qquad P = \ell_1^2 + \ell_2.$$

There are a constant $\varepsilon_P > 0$ and finite sets $A_j \subset \mathbb{R}$, with $|A_j| \to \infty$, such that

$$|P(A_j, A_j)| \le |A_j|^{2-\varepsilon_P}. \tag{8}$$

The bound holds for the sets constructed below, using the same set in both coordinates and every ordered pair in its Cartesian square. When $ab(ae - bc) \neq 0$, the polynomial is nonspecial in the sense of Elekes–Rónyai, as proved in Section 3.6.

For every sufficiently large integer ℓ, the external field theorem supplies a totally real tower F_j of growing degree d_j and $t = \lfloor (\ell - 1)^2/100 \rfloor$ distinct primes $p_i \equiv 1 \pmod 4$ splitting completely in every layer. The layers have a common root discriminant D with $\log D = O(\ell \log \ell)$ [1, Proposition 3.8(P1)–(P4)]. Let R be the product of these primes. Section 3.2 chooses ℓ for this P and then fixes the radius X.

Write $\sigma_{F_j,k} : F_j \hookrightarrow \mathbb{R}$, $1 \le k \le d_j$, for all real embeddings, and choose one of them, $\iota_j : F_j \hookrightarrow \mathbb{R}$. Define

$$\begin{aligned} B_{F_j}(X) &= \{\alpha \in \mathcal{O}_{F_j} : |\sigma_{F_j,k}(\alpha)| \le X \text{ for every } k\}, \\ A_j &= R^{-1}\iota_j\big(B_{F_j}(X)\big) \subset \mathbb{R}, \qquad n_j = |A_j| = |B_{F_j}(X)|. \end{aligned} \tag{9}$$

All conjugates of an algebraic integer must fit in the box. This selects finitely many integers because their vectors form a lattice. The chosen embedding puts these elements on the real line, and division by R scales both inputs in the same way. Both maps are injective, giving the equality of cardinalities.

The scaling allows us to count integral values before we divide by R. Define the auxiliary polynomial $P_R = \ell_1^2 + R\ell_2$. Since $\ell_i(Rx, Ry) = R\ell_i(x, y)$ for $i = 1, 2$, we have $P_R(Rx, Ry) = R^2 P(x, y)$. Thus for every $\alpha, \beta \in B_{F_j}(X)$,

$$P\left(\frac{\iota_j(\alpha)}{R}, \frac{\iota_j(\beta)}{R}\right) = R^{-2}\iota_j\big(P_R(\alpha, \beta)\big). \tag{10}$$

The auxiliary inputs and outputs are algebraic integers. The final real inputs may acquire a denominator. In the integral problem, the coefficient R makes every auxiliary output a square modulo each selected prime. We must show that enough inputs fit in the box and that the possible outputs occupy few enough places within their allowed residue classes.

3.2 Choosing the fixed arithmetic parameters

Fix the coefficients of P. The real output bound uses the coefficient sum H_P. Packing near the boundary of the output box requires the larger constant C_P. Explicitly,

$$H_P = (|a| + |b|)^2 + |c| + |e|, \qquad C_P = 2H_P + 1.$$

To choose ℓ, compare the number of primes supplied in Section 3.1 with the root discriminant of the same tower. With $d_j = [F_j : \mathbb{Q}]$, the field theorem gives

$$d_j \longrightarrow \infty, \qquad \mathrm{rd}(F_j) = D, \qquad \log D = O(\ell \log \ell). \tag{11}$$

Here D comes with the tower. The constant in the last bound is independent of ℓ and j. These joint guarantees let us compare the number of splitting primes with the discriminant of the very fields in which they split.

For the primes so supplied, form

$$R = \prod_{i=1}^{t} p_i, \qquad \rho = \prod_{i=1}^{t} \frac{p_i + 1}{2p_i}.$$

Since p_i is odd, the squares including zero occupy $(p_i + 1)/(2p_i)$ of $\mathbb{F}_{p_i}$. Thus ρ is the fraction occupied by the product of the square sets in $\prod_i \mathbb{F}_{p_i}$. This is a set of allowed outputs. The fraction of classes actually attained can be smaller. In a layer of degree d, complete splitting repeats the restriction in d coordinates for each prime, giving an allowed fraction ρ^d.

Choose ℓ sufficiently large that

$$0 < \theta := C_P D \rho < 1. \tag{12}$$

Such a choice exists for the supplied tower. Every p_i is at least 5, so each factor in ρ is at most $3/5$. Hence

$$\log(C_P D \rho) \le \log C_P + O(\ell \log \ell) + t \log(3/5).$$

The coefficient cost $\log C_P$ is fixed. The negative term has magnitude proportional to ℓ^2 for large ℓ, so it eventually exceeds the discriminant cost and the coefficient cost together. Fix one such ℓ, its tower and primes, and consequently D, R, ρ, θ.

Finally fix $X > \max\{R, \sqrt{D}\}$. The condition $X > R$ lets us bound the linear auxiliary term on the scale X^2. The condition $X > \sqrt{D}$ ensures growth of the input sets. Only the layer j and its degree d_j now vary. The estimates we need are

$$\left(\frac{X}{\sqrt{D}}\right)^{d_j} \le n_j \le (2X + 1)^{d_j}, \qquad |P(A_j, A_j)| \le \theta^{d_j} n_j^2. \tag{13}$$

3.3 Counting the inputs

Fix a layer $F = F_j$ and write $d = [F : \mathbb{Q}] = d_j$ and $n = n_j$. Under the map $\Phi_F : F \to \mathbb{R}^d$ of all real embeddings, O_F gives a lattice of covolume $\sqrt{|\Delta_F|} = D^{d/2}$. Its points in $[-X, X]^d$ correspond precisely to $B_F(X)$. Each point represents one algebraic integer with d linked conjugate coordinates.

For the lower bound, translate the half-open cube $[-X/2, X/2)^d$ through a fundamental cell of the lattice. Averaging the number of lattice points in the translated cube gives its volume divided by the

lattice covolume, namely $X^d/D^{d/2}$. Choose a translate containing at least that many points and fix one of them, v_0. The differences $v - v_0$ are distinct lattice points in $[-X, X]^d$, including zero when $v = v_0$. These differences witness the lower bound for the original centered set,

$$n = |B_F(X)| \geq (X/\sqrt{D})^d.$$

Thus $|A_j| = n_j \to \infty$, although every A_j lies in the fixed interval $[-X/R, X/R]$. The cardinality grows because the field degree grows. Squaring the same inequality will allow us to compare the output count with the actual number n_j^2 of input pairs.

For the upper bound, take two distinct elements of $B_F(X)$. Their difference γ has nonzero integer norm, so $\prod_{k=1}^{d} |\sigma_{F,k}(\gamma)| \geq 1$. At least one coordinate of their vectors differs by at least one. Cubes of side one centered at the vectors therefore have disjoint interiors, and all fit inside the cube of side $2X + 1$. Comparison of volumes gives $n \leq (2X + 1)^d$. This upper bound will turn a saving per field degree into a saving in the exponent of n.

3.4 Counting every auxiliary output

Every value $P_R(\alpha, \beta)$ satisfies two restrictions. Its reductions must lie in certain residue classes, and every real conjugate must lie in a bounded interval. We count the allowed classes and then bound, uniformly, how many distinct outputs can fit in any one of them.

For a selected prime $p \mid R$, reduction gives $P_R(x, y) \equiv (ax + by)^2 \pmod{p}$. The squares in $\mathbb{F}_p$ consist of zero and $(p - 1)/2$ nonzero values, since each nonzero square has precisely two roots of opposite sign. If either a or b is nonzero modulo p, the reduced linear form takes every value of $\mathbb{F}_p$ by varying the corresponding variable. If both coefficients vanish, its only value is zero. In either case

$$P_R(\mathbb{F}_p, \mathbb{F}_p) \subseteq \{z^2 : z \in \mathbb{F}_p\}, \qquad |P_R(\mathbb{F}_p, \mathbb{F}_p)| \leq (p + 1)/2.$$

Containment suffices even when the finite box occupies only part of the residue plane.

Complete splitting means that $p_i\mathcal{O}_F$ is a product of d distinct prime ideals with residue field $\mathbb{F}_{p_i}$. Reduction of $\mathcal{O}_F$ modulo these ideals gives d residue coordinates, with addition and multiplication component by component. The Chinese remainder theorem, first over each p_i and then across the distinct primes, gives

$$\mathcal{O}_F/R\mathcal{O}_F \cong \prod_{i=1}^{t} \mathbb{F}_{p_i}^d.$$

For $d > 1$ the direct product $\mathbb{F}_{p_i}^d$ has zero divisors, whereas the field $\mathbb{F}_{p_i^d}$ has none. Each coordinate here comes from a prime ideal of $\mathcal{O}_F$. The maps to $\mathbb{R}$ used to bound the conjugates play a separate role. Counting the product of the allowed square sets gives at most

$$\prod_{i=1}^{t} \left(\frac{p_i + 1}{2} \right)^d = (\rho R)^d \tag{14}$$

classes containing actual outputs.

To bound their real coordinates, take $\alpha, \beta \in B_F(X)$ and any real embedding $\sigma_{F,k}$. Both input coordinates have absolute value at most X, so

$$|\sigma_{F,k}(P_R(\alpha, \beta))| \leq (|a| + |b|)^2 X^2 + R(|c| + |e|)X \leq H_P X^2 =: H.$$

The last step uses $R < X$. All output vectors therefore lie in $[-H, H]^d$.

Now fix one residue class modulo $R\mathcal{O}_F$. The difference of two distinct outputs in this class is $R\gamma$ with $0 \neq \gamma \in \mathcal{O}_F$. The integer norm argument from Section 3.3 gives $\max_k |\sigma_{F,k}(\gamma)| \geq 1$. Within this class the vectors are consequently at least R apart in the maximum norm. Cubes of side R centered at them have disjoint interiors and all lie in the larger cube of side $2H + R$. Comparing volumes gives a bound of $(2H/R + 1)^d$ points in the class. The argument works for each class separately, including when $H = 0$ and this bound is one.

Multiplying the number of classes by the uniform bound in one class gives

$$\begin{aligned} |P_R(B_F(X), B_F(X))| &\leq (\rho R)^d (2H/R + 1)^d = [\rho(2H + R)]^d \\ &\leq (C_P \rho X^2)^d \leq (C_P D \rho)^d n^2 = \theta^d n^2. \end{aligned} \tag{15}$$

Indeed, $R < X^2$ and $C_P = 2H_P + 1$ give $2H + R \leq C_P X^2$. The lower input count gives $X^{2d} \leq D^d n^2$ in the final inequality. This is where the discriminant cost enters the comparison with the number of pairs. Every ordered pair in $B_F(X)^2$ yields an output in one of the classes just counted.

A local illustration uses $P_*(x, y) = (x + 2y)^2 + 3x - y$ and the illustrative modulus 5. The auxiliary polynomial $(P_*)_5 = (x + 2y)^2 + 5(3x - y)$ takes precisely the residues 0, 1, 4 on $\mathbb{F}_5^2$, and the common scaling gives $P_*(\alpha/5, \beta/5) = (P_*)_5(\alpha, \beta)/25$. Here $C_{P_*} = 27$, so the single factor $3/5$ does not pay even the coefficient cost. The global argument uses the primes selected together in Section 3.2.

3.5 The power saving for the original polynomial

Take the sets A_j from (9). As α, β range independently over $B_{F_j}(X)$, the scaled inputs in (10) range over all of $A_j \times A_j$. We therefore have an equality of sets,

$$P(A_j, A_j) = R^{-2} \iota_j \Big(P_R(B_{F_j}(X), B_{F_j}(X)) \Big).$$

The embedding and the nonzero output scaling preserve the number of distinct values, including when many pairs have the same polynomial value. By (15), $|P(A_j, A_j)| \leq \theta^{d_j} n_j^2$.

The upper input count yields $d_j \geq \log n_j / \log(2X + 1)$. Since $0 < \theta < 1$, increasing its exponent decreases the result. Hence

$$\theta^{d_j} \leq \theta^{\log n_j / \log(2X+1)} = n_j^{-\varepsilon_P}, \qquad \varepsilon_P = \frac{-\log \theta}{\log(2X + 1)} > 0. \tag{16}$$

This is the degree-to-cardinality comparison of Section 2.3 with $\gamma = -\log \theta$ and $B = \log(2X + 1)$. Both constants were fixed before the degree grew. Together with $n_j \to \infty$, the exact bounds prove Theorem 1 with the displayed ε_P.

The proof also gives an interpretation in terms of repetition. Dividing the n_j^2 input pairs by the number of attained outputs shows that the mean fiber size is at least $n_j^{\varepsilon_P}$. This average permits individual fibers to have fewer pairs.

3.6 Scope of the theorem and the nonspecial subfamily

The image bound applies to every fixed integer quadruple in the theorem. To see which examples connect directly to the Elekes–Rónyai exceptions of Section 1.2, compare the two matrices

$$\begin{pmatrix} a & b \\ c & e \end{pmatrix}, \qquad \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}.$$

The first has rows given by ℓ_1 and ℓ_2. Its determinant $ae - bc$ tests whether these forms are independent. The second represents the homogeneous quadratic part ℓ_1^2, the sum of the terms of degree two. Its determinant is always zero. Its rank is one when $(a, b) \neq (0, 0)$ and zero when $a = b = 0$. For $P_* = (x + 2y)^2 + 3x - y$, the rows of the first matrix are $(1, 2)$ and $(3, -1)$, giving determinant -7. The quadratic matrix has rows $(1, 2)$ and $(2, 4)$, giving determinant zero.

Suppose now that $ab(ae - bc) \neq 0$. We show that neither exceptional representation is possible [10, Lemma 4.4]. In an additive representation $P = h(f(x) + g(y))$, dependence on both variables forces h, f, g to be nonconstant. If h were linear, there would be no mixed xy term, although P has coefficient $2ab \neq 0$. The total degree is $\deg h \max\{\deg f, \deg g\} = 2$, since the leading pure powers in different variables cannot cancel. Thus h is quadratic and f, g are affine of degree one. Write $f(x) + g(y) = U(x, y) + k$ with U homogeneous linear. The degree-two part of $h(U + k)$ is a nonzero scalar multiple of U^2 and equals ℓ_1^2. The two linear forms vanish on the same line, so U is proportional to ℓ_1. The degree-one part of $h(U + k)$ is also a multiple of U. This would make ℓ_2 proportional to ℓ_1, contradicting $ae - bc \neq 0$.

In a multiplicative representation $P = h(f(x)g(y))$, both variables again occur. The degree equation $\deg h\,(\deg f + \deg g) = 2$ forces all three polynomials to have degree one. Their composition then has a quadratic part proportional to xy. In ℓ_1^2, both the x^2 and the y^2 coefficients are nonzero. This excludes the multiplicative form and proves nonspecialness under the stated condition.

If $ab \neq 0$ and $ae - bc = 0$, then $\ell_2 = \lambda\ell_1$ for a scalar λ. Here $P = h(\ell_1)$ with $h(t) = t^2 + \lambda t$, an additive exceptional form. If $a = b = 0$, then $P = cx + ey$, possibly zero. An interval of n consecutive integers gives at most $(|c| + |e|)(n - 1) + 1$ output integers. If exactly one of a, b vanishes, both variables can still occur, as in $x^2 + y$. All these cases are included in the construction.

Corollary (Rational rank-one quadratics). Let $f \in \mathbb{Q}[x, y]$ be a fixed quadratic polynomial whose homogeneous quadratic part has rank one. There are a constant $\varepsilon_f > 0$ and finite sets $A_j \subset \mathbb{R}$, with $|A_j| \to \infty$, such that

$$|f(A_j, A_j)| \leq |A_j|^{2-\varepsilon_f}.$$

Proof. Choose a positive integer m clearing all denominators and write

$$\widetilde{f} = mf = \alpha x^2 + \beta xy + \gamma y^2 + \delta x + \eta y + \kappa, \qquad \alpha, \beta, \gamma, \delta, \eta, \kappa \in \mathbb{Z}.$$

The quadratic part still has rank one, so $\beta^2 = 4\alpha\gamma$ and $(\alpha, \beta, \gamma) \neq (0, 0, 0)$. If $\alpha \neq 0$, this identity gives

$$P(x, y) := 4\alpha\big(\widetilde{f}(x, y) - \kappa\big) = (2\alpha x + \beta y)^2 + 4\alpha\delta x + 4\alpha\eta y.$$

Thus P belongs to the integer-coefficient family in Theorem 1. For every finite $A \subset \mathbb{R}$,

$$P(A, A) = 4\alpha m\, f(A, A) - 4\alpha\kappa.$$

The output map $t \mapsto 4\alpha mt - 4\alpha\kappa$ has nonzero slope, so it preserves cardinality. Hence the sets and exponent supplied by Theorem 1 for P also work for f.

If $\alpha = 0$, rank one gives $\beta = 0$ and $\gamma \neq 0$. Apply the preceding argument to $f(y, x)$, whose x^2 coefficient after clearing denominators is γ. Interchanging coordinates permutes $A \times A$, so $\{f(y, x) : x, y \in A\} = f(A, A)$. □

Section 3 therefore gives a fixed power saving for the entire image of every fixed rational-coefficient quadratic with rank-one homogeneous part, on suitable finite real Cartesian squares of arbitrarily large size. The same set is used in both inputs, and every ordered pair is included. The full-rank quadratics and higher-degree targets studied in Section 4 require additional analysis.

4 Extensions, obstructions and unresolved questions

The square leading term in Theorem 1 restricts every output modulo the selected primes, while quadratic growth keeps the real output region small enough to count. The next constructions separate these two advantages. Full-rank quadratic forms preserve the degree but change the local images. Higher powers can retain sparse local images while enlarging the real output region.

4.1 Full-rank quadratics: norms, shifts and a transfer theorem

At good primes a full-rank homogeneous quadratic represents every residue, and removing a linear term can require different shifts in the two inputs. For an integral quadratic form $q(x, y) = Ax^2 + Bxy + Cy^2$, full rank means that its symmetric matrix $\left(\begin{smallmatrix} A & B/2 \\ B/2 & C \end{smallmatrix}\right)$ has nonzero determinant. For this full-rank form a good prime is an odd prime at which the determinant remains nonzero. On the full plane $\mathbb{F}_p^2$, a nonzero rank-one form represents $(p + 1)/2$ residues, while a full-rank form represents all p residues [10, Lemma 4.2]. In the latter case the form is equivalent either to a scaled product uv or to a scaled norm from $\mathbb{F}_{p^2}$. The norm is onto on nonzero elements, and zero is also represented. Any attempt to recover a density saving by restricting the inputs must also count the pairs retained.

A norm gives another way to make values repeat. Consider $Q_\omega(x, y) = x^2 + xy + y^2$ over a characteristic-zero field k. Write ω for a primitive cube root of unity, so $\omega^3 = 1$, $\omega \neq 1$ and $\omega^2 + \omega + 1 = 0$. If $E_k = k(\omega)$ is quadratic over k, its involution τ interchanges ω and ω^2, and its relative norm is $N(z) = N_{E_k/k}(z) = z\tau(z)$. Hence

$$Q_\omega(x, y) = (x - \omega y)(x - \omega^2 y) = N(x - \omega y).$$

Multiplicativity gives $N(uz) = N(z)$ whenever $N(u) = 1$. Section 4.2 examines which of these symmetries act on actual pairs of inputs.

Adding a linear term can cause a different problem. For the benchmark $P_0(x, y) = Q_\omega(x, y) + x$, expansion gives

$$\begin{aligned} P_0(x, y) &= Q_\omega(x + 2/3, y - 1/3) - 1/3, \\ P_0(A, A) &= Q_\omega(A + 2/3, A - 1/3) - 1/3. \end{aligned} \tag{17}$$

A bound for a homogeneous image on $C \times C$ gives no immediate control over these unequal translates. The general characteristic-zero reduction and the role of the trace are given in Appendix C.1.

The transfer in [11, §2.3] resolves the mismatch under two hypotheses. Suppose $C \subset \mathbb{R}$ is finite, $N = |C| \geq 2$, and

$$|Q_\omega(C, C)| \leq K_0 N^{2-c_0}, \qquad |C - C| \leq D_0 N^{1+\eta}, \qquad C - C = \{u - v : u, v \in C\},$$

where $K_0, D_0 > 0$, $0 < c_0 \le 1$ and $0 \le \eta < c_0/2$ are fixed. The construction takes

$$B = C \cap (C - h), \qquad A = B/h + 1/3,$$

where h maximizes $|C \cap (C-h)|$ among the nonzero differences in $C - C$. Indeed, the multiplicities of nonzero differences sum to $N(N-1)$, so $|B| \ge N(N-1)/|C-C| \ge N^{1-\eta}/(2D_0)$. For $x = b/h + 1/3$ and $y = b'/h + 1/3$ in A, the shifted arguments are $(b+h)/h$ and b'/h. Both numerators lie in C. By homogeneity, (17) gives $P_0(A, A) \subseteq h^{-2} Q_\omega(C, C) - 1/3$. Since $|A| = |B|$, we have

$$|P_0(A, A)| \le K_0 (2D_0)^{2-c_1} |A|^{2-c_1}, \qquad 2 - c_1 = \frac{2 - c_0}{1 - \eta}, \qquad c_1 = \frac{c_0 - 2\eta}{1 - \eta} > 0.$$

Along a growing family, every fixed $0 < \varepsilon < c_1$ absorbs the constant. The implication is proved. Constructing such sets C of unbounded cardinality remains open in this project.

4.2 Norm-one units: population bounds and the return problem

The candidate for Q_ω uses algebraic integers with a common bound on all their conjugates. Take the specified totally real tower F_j of degree $d_j \to \infty$ and bounded root discriminant, let s be the real root of $s^3 + s + 1 = 0$, and put

$$L_j = F_j(s), \qquad E_j = L_j(\omega), \qquad \mathcal{R}_j = \mathcal{O}_{L_j}[\omega].$$

The order $\mathcal{R}_j$ is a subring of finite index in $\mathcal{O}_{E_j}$. For the fixed radius T_* supplied in [12, §§2–3], the actual inputs are

$$C_j = \{x \in \mathcal{O}_{L_j} : |\sigma(x)| \le T_* \text{ for every embedding } \sigma\}, \qquad \widetilde{A}_j = \iota_{L_j}(C_j) \subset \mathbb{R},$$

where $\iota_{L_j} : L_j \hookrightarrow \mathbb{R}$ is one real embedding. It preserves cardinality, and $|C_j|$ has exponential lower and upper bounds in d_j. Appendix C.2 gives the constants and field calculation.

The field L_j has d_j real places and d_j conjugate pairs of complex embeddings, hence degree $3d_j$. The quadratic extension E_j has degree $6d_j$ and no real embeddings. Its involution τ_j fixes L_j and exchanges ω, ω^2. Over real places it acts as complex conjugation. Over a nonreal place, the two absolute values of a relative norm-one unit need only be reciprocal. This allows distortion that did not arise for unit-distance directions in $F_j(i)$.

Let $\mathcal{G}_j = \{u \in \mathcal{R}_j^\times : u\tau_j(u) = 1\}$. Writing $u = \alpha + \beta\omega$ and multiplying $x - \omega y$ gives a matrix of determinant $\alpha^2 - \alpha\beta + \beta^2 = u\tau_j(u) = 1$. The action and its inverse preserve $\mathcal{O}_{L_j}^2$ because both matrices have entries in $\mathcal{O}_{L_j}$,

$$M_u = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha - \beta \end{pmatrix}, \qquad M_u^{-1} = \begin{pmatrix} \alpha - \beta & -\beta \\ \beta & \alpha \end{pmatrix}.$$

Multiplicativity of the norm gives $Q_\omega(M_u \mathbf{p}) = Q_\omega(\mathbf{p})$. A *return* from $\mathbf{p} \in C_j^2$ also requires $M_u \mathbf{p} \in C_j^2$.

Which units might return? For each of the d_j complex places of L_j, choose one extension v_i^+ to E_j and set $\mathrm{Log}_j(u)_i = \log |v_i^+(u)|$. The finite set of candidates is

$$U_j = \left\{ u \in \mathcal{G}_j : \sum_i |\,\mathrm{Log}_j(u)_i| \le d_j H_{\mathrm{cut}}(T_*) \right\}, \qquad H_{\mathrm{cut}}(T) = \tfrac{1}{2} \log(48T^6).$$

The population theorem gives $|U_j| \geq 2^{d_j}$ at every level [12]. A separate norm argument places every unit returning a nonzero input in U_j (Appendix C.2). The cutoff bounds total logarithmic size and permits individual conjugates to grow with d_j. The units 1 and -1 give two distinct returning targets at every nonzero source. An exponential lower bound for the mean number of actual returns remains open.

Each orbit of the full norm-one order-unit group $\mathcal{G}_j$ on the ambient algebraic pairs has a single quadratic output value. Counting the nonzero orbits meeting C_j^2 and adding the value at the origin therefore bounds the entire image. For $z = x - \omega y \neq 0$, its orbit is determined by the pair $(z\mathcal{R}_j, z\tau_j(z))$, where $z\mathcal{R}_j = \{zr : r \in \mathcal{R}_j\}$. Equal principal ideals give a unit ratio and equality of the norm values makes that unit have relative norm one. Conversely, multiplication by such a unit preserves both labels. Let o_j count the nonzero orbits meeting C_j^2. If an orbit meets that square in m points, exhaustion of returns gives exactly $T_j(\mathbf{p}) = m$ distinct targets under U_j for each point in the intersection. Thus [13, incidence companion, §§2–4]

$$\sum_{\mathbf{p} \in C_j^2 \setminus \{(0,0)\}} \frac{1}{T_j(\mathbf{p})} = o_j, \qquad |Q_\omega(\widetilde{A}_j, \widetilde{A}_j)| \leq 1 + o_j. \tag{18}$$

Each orbit intersection of size m contributes $m(1/m) = 1$ to the sum. The origin accounts for the extra value. Different orbits can share one quadratic value. A few large orbit intersections can dominate the mean return count over nonzero input pairs, whereas the reciprocal sum counts each occupied orbit once. A bound making $o_j/|C_j|^2$ exponentially small in d_j would suffice for compression. Neither the population theorem nor the orbit identity establishes this bound.

A different family built from cyclotomic factors and their rotations has a proved large image. Appendix C.3 defines its real sets A_r and explains why the full image satisfies $|Q_\omega(A_r, A_r)| \geq |A_r|^2/(18M)$ for one fixed M at all sufficiently large depths [15]. This excludes fixed power saving on this family, including its growing subsequences. The relative-unit sets $\widetilde{A}_j$ above are separate.

4.3 A quartic test: local savings and the limits of interval packing

For the fixed quartic $P_4(x, y) = (x + 2y)^4 + 3x - y$, restricting the inputs at the prime 7 gives a local density saving too small for the global interval-packing estimate below. The analogue of the quadratic construction uses

$$(P_4)_R(x, y) = (x + 2y)^4 + R^3(3x - y), \qquad (P_4)_R(Rx, Ry) = R^4 P_4(x, y).$$

A prime dividing R leaves fourth powers on reduction. Their density depends on the prime. The difficulty at the real embeddings is that an input radius X now gives fourth-degree growth, against only a square of the input population. The local construction below quantifies how much density can be gained. Appendix C.4 also treats the attempted combination of units and additive inputs.

The prime 7 is singled out by the linear forms $x + 2y$ and $3x - y$, whose coefficient determinant is -7. In fact $(x + 2y) + 2(3x - y) = 7x$. In the source's totally ramified cubic extension of $\mathbb{Q}_7$, let $\mathcal{O}_{\mathrm{loc}}$ be the valuation ring. Its maximal ideal $\mathfrak{p}$ has field quotient $\mathcal{O}_{\mathrm{loc}}/\mathfrak{p} = \mathbb{F}_7$. The identity $7\mathcal{O}_{\mathrm{loc}} = \mathfrak{p}^3$ specifies ramification index three. The finite ring $S_m = \mathcal{O}_{\mathrm{loc}}/\mathfrak{p}^{3m}$ has 7^{3m} elements and characteristic 7^m. Under the reduction $S_m \to \mathbb{F}_7$, let B_m^{loc} be the entire preimage of any fixed $\alpha \in \{3, 5, 6\}$. It occupies $1/7$ of S_m.

The local theorem gives 4 outputs at $m = 1$ and $22 \cdot 7^{3m-4}$ at every $m \geq 2$ [16, §§1–4]. At higher precision the output fraction is $22/7^4$. Dividing by the pair-input fraction $(1/7)^2$ gives the

normalized factors

$$\frac{|P_4(B_m^{\mathrm{loc}}, B_m^{\mathrm{loc}})|/|S_m|}{(|B_m^{\mathrm{loc}}|/|S_m|)^2} = \begin{cases} 4/7, & m = 1, \\ 22/49, & m \geq 2. \end{cases}$$

The stabilization has an algebraic explanation. Three complete classes modulo $\mathfrak{p}^3$ and one modulo $\mathfrak{p}^4$ occur, and an independent input fills every higher digit (Appendix C.4).

The global test takes a specific totally real tower of degree d, with $d/3$ local factors of degree three at 7, and a translate of an ideal lattice with input radius X. Each real coordinate lies in $I = [-1 - X, -1 + X]$. Write $W = \max_{I^2} P_4 - \min_{I^2} P_4$ for the full continuous width. For $m \geq 2$ the sufficient logarithmic saving per global degree is

$$\kappa_m = \log \frac{X^2}{D(22/49)^{1/3}(W + 7^m)},$$

where D is the tower's root discriminant. The cube root comes from having $d/3$ local factors in degree d. On the diagonal $f(t) = P_4(t, t) = 81t^4 + 2t$. With $b = -1$, the average of $f(b + X)$ and $f(b - X)$ minus $f(b)$ is $81X^4 + 486b^2X^2$, so $W \geq 81X^4$. Consequently

$$(W + 7^m)/X^2 \geq 81X^2 + 7^m/X^2 \geq 18\,7^{m/2}.$$

The last inequality is arithmetic-geometric mean. It gives $\kappa_m < 0$, whereas this method needs a positive rate. The tested finer covers also fail when assigned the full continuous intervals and the same uniform packing estimates [16, §§5–6]. A discrete image can occupy far less than its containing interval, so this failure gives no lower bound for the actual quartic image.

4.4 Cubic tests: quadratic transfer, rich fibers and digit sets

Consider the two targets $P_+(x, y) = (x - y)^3 + x + y$ and $P_-(x, y) = (x - y)^3 - 3(x + y)$. The first test asks whether inputs that compress the quadratic $P_2 = (x - y)^2 + 5(x + y)$ also compress P_-. The two polynomials satisfy

$$5P_- = 5(x - y)^3 + 3(x - y)^2 - 3P_2.$$

Knowing the quadratic value still leaves the difference undetermined. For instance, $(1, 0)$ and $(0, 1)$ both give $P_2 = 6$, but their P_- values are -2 and -4. For the following family this loss of collisions persists at arbitrarily large sizes.

Take the transfer construction's own totally real fields F_j, with $d_j = [F_j : \mathbb{Q}] \to \infty$ and $\mathrm{rd}(F_j) \leq D$. Fix distinct rational primes splitting completely in every field and let Q be their product. Write $B_{F_j}(Q/8)$ for the algebraic integers whose absolute values at all real embeddings are at most $Q/8$. With a real embedding ι_j, put $A_j = Q^{-1}\iota_j(B_{F_j}(Q/8))$. The condition $Q/8 > \sqrt{D}$ gives growth by the same centered-box lower bound as in the quadratic proof.

When every selected prime satisfies $q \equiv 5 \pmod{12}$, P_- is injective on A_j^2 [17, quadratic-to-cubic checkpoint, §D.3]. To see this, write a pair by its algebraic numerators a, b and set $u = a - b$, $v = a + b$. Clearing the denominator gives $u^3 - 3Q^2v$. Equal values give equal cubes modulo each prime-ideal factor above q. Since $q \equiv 2 \pmod 3$, cubing permutes $\mathbb{F}_q$, and complete splitting and CRT give equality of the differences modulo $Q\mathcal{O}_{F_j}$. The difference of two candidates for u has absolute value at most $Q/2$ in every embedding. Dividing it by Q gives an algebraic integer all of whose conjugates have absolute value less than one, so it vanishes. The cubic value then determines v and hence $(a, b) = ((v + u)/2, (v - u)/2)$.

For $q \equiv 1 \pmod{12}$ at every selected prime, the source instead bounds the off-diagonal collisions. Under its successful quadratic estimate (Appendix C.5), the result is $|P_-(A_j, A_j)| \geq |A_j|^2/(1+e^{-\delta d_j})$ for a fixed $\delta > 0$. Both cases concern the specified quadratic witnesses.

The next construction concentrates collisions in a single fiber of P_- while producing an almost maximal image. Define $x(i) = 36i^3+3i+2$, $y(i) = 36i^3-3i+2$ and $A_N = \{x(i_j), y(i_j) : 1 \leq j \leq N\}$. The rich-fiber checkpoint chooses increasing positive integers i_j by finite avoidance. At each step only finitely many choices create an unintended equality between outputs. For the resulting $2N$ inputs the N designated pairs all give $P_-(x(i_j), y(i_j)) = -12$, and every other ordered pair gives a distinct value. Merging exactly N pairs into one value leaves $4N^2 - N + 1$ outputs [17, rich-fiber checkpoint, §§B–C, Theorem 3]. Appendix C.5 explains why the avoidance works.

For P_+, the digital investigation reports an obstruction for digit sets with fixed prime support. Fix a nonempty finite set of primes S and $\alpha > 0$. There is an $\eta(S, \alpha) > 0$ such that, for every sufficiently large S-smooth integer N and every $\mathcal{D} \subseteq \{0, \ldots, N-1\}$ of size at least N^α, the cleared integer polynomial satisfies

$$Z_N(U, V) = (U - V)^3 + N^2(U + V), \qquad |Z_N(\mathcal{D}, \mathcal{D})| \geq \eta(S, \alpha)|\mathcal{D}|^2.$$

Here S-smooth means that all prime factors lie in S, for example $N = 2^i 3^j$ when $S = \{2, 3\}$. Since $N^3 P_+(U/N, V/N) = Z_N(U, V)$, the stated theorem excludes a fixed saving on these digit sets. The T01 state records the theorem as proved [17, T01, §4.3, (T01.6)]. Its full proof was unavailable for this report. The statement requires fixed prime support and positive-power input growth.

4.5 Adaptive radical sets and the missing growth estimate

This construction seeks a power saving for $Q_\omega(x, y) = x^2 + xy + y^2$ by choosing successive radical lifts. For a finite nonempty symmetric set $A = -A \subset \mathbb{R}$, put $M = \max_{a \in A} |a|$. The radical construction chooses $C > M$ and forms

$$\mathrm{Rad}_C(A) = \left\{ \pm\sqrt{(C + a)/(C - M)} : a \in A \right\}.$$

The positive roots increase strictly with a and are nonzero. Their negatives are disjoint from them, so the number of inputs doubles and the number of ordered pairs grows by a factor of four. The question is whether repeated choices of C can make the number of *distinct* quadratic outputs grow more slowly.

Start at $A_0 = \{-1, 1\}$. At each step choose C to minimize the next image over a finite list of all legal parameters producing collisions and the fallback $M + 1$, with ties settled by the least listed parameter. Appendix C.6 gives the exact collision polynomial and the finite selection rule. This minimizes the next cardinality only. The exact levels recorded in [14, §§3–4.2] are

Level k	0	1	2	3	4
Input cardinality $\lvert A_k \rvert$	2	4	8	16	32
Quadratic image cardinality s_k	2	5	17	68	269

The chosen parameters taking level 0 through level 4 are $5/3, 5/2, 4, 3$. The next optimizer is unknown in the cited record.

Since $n_k = |A_k| = 2^{k+1}$, a bound $s_k \leq C_1 \lambda^k$ for fixed $C_1 > 0$ and $1 < \lambda < 4$ would give $s_k \leq (C_1/\lambda) n_k^{\log_2 \lambda}$. Every fixed $0 < \varepsilon < 2 - \log_2 \lambda$ would absorb the constant at large levels. The same bound along an unbounded subsequence would already suffice. The five counts do not establish such a rate. Appendix C.6 gives the specific signed-root estimate that remains open.

4.6 Summary of results and open problems

Theorem 1 gives a fixed power saving on arbitrarily large Cartesian squares for every fixed integer polynomial of the form $(ax + by)^2 + cx + ey$. The condition $ab(ae - bc) \neq 0$ identifies its nonspecial subclass. The corollary in Section 3.6 extends the compression result to every fixed rational-coefficient quadratic whose homogeneous quadratic part has rank one. For full-rank forms, the proved shift transfer still needs a witness satisfying two bounds, while the relative-unit population theorem leaves actual returns to be counted. Some tested families provably have large images. In contrast, the quartic interval calculation limits one upper-bound method, and the radical data establish exact finite counts. None of these extensions yet supplies a complete construction for its fixed target.

Appendix A. Notation and coordinate conventions

A field subscript specifies which embeddings or norm are being used. Within one construction the index j may be suppressed. Each occurrence of d_j below refers to the degree of the totally real base field in that construction. A signature (r_1, r_2) lists the numbers of real embeddings and of conjugate pairs of complex embeddings. The degree is $r_1 + 2r_2$.

Field	Degree	Signature	Role
F_j	d_j	$(d_j, 0)$	Totally real base for the construction in its section.
$K_j = F_j(i)$	$2d_j$	$(0, d_j)$	Unit-distance directions. The involution c fixes F_j and sends i to $-i$.
$L_j = F_j(s)$	$3d_j$	(d_j, d_j)	The mixed-signature field of the unit-return construction.
$E_j = L_j(\omega)$	$6d_j$	$(0, 3d_j)$	The relative involution τ_j fixes L_j and exchanges ω, ω^2.

Notation	Meaning in this report
$\sigma_{F,k}$, Φ_F	One real embedding, and the map to $\mathbb{R}^d$ recording them all.
$\sigma_{K,k}$, Φ_K, π_1	One complex embedding above the k-th real place, the map to $\mathbb{C}^d$ selecting one embedding above each real place, and projection to its first coordinate.
ι_j	A single embedding into $\mathbb{R}$ for the indicated field.
$N_{H/H'}$, Log	Multiplicative field norm to the specified subfield, and the logarithmic absolute-value map with the normalization of its section.
Q, r, $\mathbf{a}$	Denominator parameter, disc radius and ambient translate in the planar construction.
ℓ_1, ℓ_2, R, ρ	Two integer linear forms, the squarefree modulus and the fraction of allowed square residues in the polynomial construction.
Q_ω, $\mathcal{R}_j$	The form $x^2 + xy + y^2$ and the order $\mathcal{O}_{L_j}[\omega]$.
$T_j(\mathbf{p})$, o_j	Number of returns of the indicated pair and number of occupied nonzero group orbits.
S_m, B_m^{loc}	The quartic local quotient ring and its restricted input.
$\mathrm{Rad}_C(A)$	The radical input lift. The T03 source writes $\mathcal{L}_C(A)$ for this lift and reserves $\mathcal{T}_C(A)$ for signed outputs.

The determinant of a lattice basis measures volume in its specified coordinates. The determinant $ae - bc$ tests independence of two linear forms. The determinant of a symmetric quadratic matrix tests the rank of its quadratic part in dimension two. Section 3.6 computes the latter two determinants for the same polynomial.

Appendix B. Supporting arithmetic and counting arguments

B.1 Split-prime towers and norm-one directions

An ideal in a ring is an additive subgroup closed under multiplication by ring elements. Two nonzero fractional ideals in K represent the same ideal class when their quotient is principal, that is, generated by a single element of $K^\times$. There are $h(K)$ such classes. In Section 2.1 this finite number measures the loss in converting many choices of ideals into direction elements.

Here is the field input in the notation of the report. For each sufficiently large integer ℓ, set $t = \lfloor(\ell-1)^2/100\rfloor$. Proposition 3.8 of [1] supplies a totally real cyclic cubic field F_0, an infinite tower of everywhere-unramified Galois extensions F_j/F_0 of 3-power degree, and t distinct rational primes $q_b \equiv 1 \pmod 4$ splitting completely in all the F_j. The layers have growing degrees d_j and constant root discriminant $\mathrm{rd}(F_0)$, with $\log \mathrm{rd}(F_0) = O(\ell \log \ell)$. For $K_j = F_j(i)$ the source also bounds $h(K_j) \le H_\ell^{d_j}$ with $\log H_\ell = O(\ell \log \ell)$. Since t grows quadratically in ℓ, one can fix ℓ so that $\gamma = t \log 2 - \log H_\ell > 0$. The primes and $Q = \prod_b q_b$ are then fixed at all levels.

The tower proof arranges both unlimited degree and complete splitting. In the source's construction a large elementary abelian unramified extension gives many generators for the group of the maximal unramified pro-3 extension. A Frobenius element records the action of a prime on residue fields in an unramified extension. Trivial Frobenius in a finite Galois layer gives complete splitting there. Chebotarev's theorem selects rational primes splitting in an auxiliary *finite* extension, so their Frobenius representatives lie in the Frattini subgroup of the pro-3 group. Imposing their triviality leaves the minimal generator count unchanged. There are at most $3t$ added relations, one per prime above the chosen rational primes in the cubic base. The source's relation bound then meets the Golod–Shafarevich criterion for the quotient to remain infinite. Finite layers of that quotient give growing degrees and keep the chosen primes split. The arithmetic inputs and the proof of the relation estimates are in [1, Propositions 3.2–3.8]. The companion paper gives a separate treatment of the arithmetic construction [2].

Fix one layer F and write $K = F(i)$. Each q_b has $d = [F : \mathbb{Q}]$ prime factors in F. Over each such factor $x^2 + 1$ has two distinct roots in $\mathbb{F}_{q_b}$, so in K there are td pairs $\{\mathfrak{p}_s, c(\mathfrak{p}_s)\}$ permuted by conjugation. A binary vector $v \in \{0,1\}^{td}$ chooses one factor from each pair, with $v_s = 1$ selecting $\mathfrak{p}_s$. Let I_v be their product. Unique ideal factorization distinguishes all 2^{td} products. In some ideal class there are at least $2^{td}/h(K)$ of them. Fix one member I_w of this class and choose α_v generating I_v/I_w for each member, taking $v = w$ with $\alpha_w = 1$.

Set $u_v = \alpha_v / c(\alpha_v)$. Its relative norm is one. In the ideal (u_v) the exponent of $\mathfrak{p}_s$ is $2(v_s - w_s)$ and the exponent of its conjugate is the negative of this integer. Indeed, division by I_w first subtracts the two choice vectors, and division by the conjugate then subtracts the opposite difference. The resulting exponents distinguish the v's. Each prime above Q occurs to exponent one in (Q), so multiplication by Q^2 clears all negative exponents. We have constructed distinct elements in $Q^{-2}O_K$ and retained at least $2^{td}/h(K) \ge e^{\gamma d}$ of them [1, Proposition 2.2].

B.2 Complex embeddings and lattice covolume

In the unit-distance construction $K = F(i)$ has degree $2d$. One representative of each conjugate pair gives $\Phi_K \colon K \to \mathbb{C}^d$. In each coordinate use ordinary area in the real and imaginary parts. To obtain the two conjugate rows of the full embedding matrix of an integral basis from the corresponding pair of real rows, one sends (x, y) to $(x + iy, x - iy)$. This two-row change of basis has determinant $-2i$. The absolute determinant of the full embedding matrix is $\sqrt{|\Delta_K|}$ by the definition of the discriminant. Removing the d factors of 2 yields

$$\mathrm{covol}\big(\Phi_K(O_K)\big) = 2^{-d}\sqrt{|\Delta_K|}.$$

Scaling each of $2d$ real coordinates by Q^{-2} multiplies this covolume by Q^{-4d}. This is the denominator in each of the two averages in Section 2.1, so its numerical value cancels when the averages are compared. The packing argument uses individual norm separation and does not need an asymptotic estimate of lattice points in each translate.

B.3 Lattice counts for algebraic integers and units

In a totally real field F of degree $d \ge 2$, the additive embedding lattice has covolume $\sqrt{\Delta_F}$. For $T \ge 1$, the source's count is

$$\frac{T^d}{\sqrt{\Delta_F}} \le |B_F^+(T)| \le (2T+1)^d \qquad \text{[3, Lemma 3.3].}$$

For the lower bound some translate of $[-T/2, T/2)^d$ contains at least its volume divided by the covolume in lattice points. Subtract one of those points from each point, including itself. All resulting differences are distinct and lie in the centered box. The upper bound packs side-one cubes about the integral points using norm separation.

The unit lattice is the image of $O_F^\times$ under $\mathrm{Log}_F(g) = (\log|\sigma_{F,k}(g)|)_{k=1}^d$ in the hyperplane $H_0 = \{z : \sum_k z_k = 0\}$. Dirichlet's unit theorem gives rank $d-1$ for this lattice. The logarithmic map has kernel $\{1,-1\}$. The usual regulator is the covolume after deleting one coordinate. Restoring the last coordinate by $z_d = -\sum_{k<d} z_k$ has Gram matrix $I + \mathbf{1}\mathbf{1}^{\mathsf{T}}$ of determinant d, so Euclidean volume in H_0 gives covolume $\sqrt{d}\ \mathrm{Reg}(F)$. For example, for $\mathbb{Q}(\sqrt{2})$ the standard fundamental unit $1+\sqrt{2}$ has logarithmic vector $(h,-h)$ with $h = \log(1+\sqrt{2})$. The ordinary regulator is h and the Euclidean length of this fundamental segment is $\sqrt{2}h$.

The actual box count, including both unit signs, satisfies for $Y \ge 1$

$$\frac{Y^{d-1}}{\sqrt{d}\ \mathrm{Reg}(F)} \le |B_F^\times(Y)| \le 10(5Y+1)^{d-1} \qquad \text{[3, Lemma 3.5].}$$

The lower estimate uses a translate of the half-size box in H_0 and the lower bound on its section volume. Taking differences of the log vectors corresponds to dividing units. For the upper estimate, distinct log vectors are separated in maximum norm by at least $\log\varphi$ by the unit-separation lemma. Packing within the hyperplane, with the section-volume upper bound, gives the stated estimate. The section-volume inequality is an imported geometric result [3, Eq. (3.1)].

These counts explain the powers of Δ_F in Section 2.2. The additive input lower bound incurs $\Delta_F^{1/2}$ in its denominator. The unit lower bound incurs $\mathrm{Reg}(F) \le \Delta_F$ by [3, Lemma 3.1]. Multiplying them produces the $\Delta_F^{3/2}$ cost for the number of inputs, with the fixed factor η^d and the factor $d^{-1/2}$ absorbed by an absolute constant to the power d. For products, $|G|$ occurs twice in the denominator of the normalized comparison, giving the possible Δ_F^2 cost. This is why the estimates must be compared after fixing a bounded root discriminant. The project reproduction develops these steps separately [5, §§4–11].

B.4 What collision energy measures

For a nonempty finite set A of size n, the collision energy $\mathcal{E}_P(A) = \sum_{z\in P(A,A)} r_P(z)^2$ counts the ordered quadruples $(a,b,a',b') \in A^4$ for which $P(a,b) = P(a',b')$. All n^2 ordered pairs occur in the partition into fibers. By Cauchy–Schwarz,

$$n^4 = \left(\sum_{z\in P(A,A)} r_P(z)\right)^2 \le |P(A,A)|\,\mathcal{E}_P(A), \qquad |P(A,A)| \ge \frac{n^4}{\mathcal{E}_P(A)}.$$

Increasing the energy weakens this particular lower bound. For an upper bound the reciprocal identity (6) describes exactly which multiplicities matter. For example, if *every* attained value has at least m representations, then summing at most $1/m$ over each input pair gives $|P(A,A)| \le n^2/m$. Knowledge of one rich fiber leaves the other terms undetermined.

Appendix C. Technical details for the extension attempts

C.1 Completing a quadratic norm and translating its inputs

Over a characteristic-zero field k, consider $q(x, y) = ax^2 + bxy + cy^2$ with $\Delta = b^2 - 4ac \neq 0$. If $a \neq 0$, put $E = k[t]/(t^2 - \Delta)$ and $\alpha = (-b + t)/(2a)$. The involution τ sends t to $-t$. This quadratic *étale algebra* is a field when Δ is nonsquare and $k \times k$ when Δ is square. In both cases $\alpha + \tau(\alpha) = -b/a$ and $\alpha\tau(\alpha) = c/a$, so $q = aN(x - \alpha y)$ with $N(z) = z\tau(z)$.

For $q + dx + ey + r$, set $z = x - \alpha y$ and $\beta = d/2 + (bd - 2ae)t/(2\Delta)$. Writing $\mathrm{Tr}(w) = w + \tau(w)$ gives $\mathrm{Tr}(\beta z) = dx + ey$. Expanding the norm now yields

$$aN(z) + \mathrm{Tr}(\beta z) + r = aN\big(z + \tau(\beta)/a\big) + r - N(\beta)/a.$$

If $a = 0$, full rank gives $b \neq 0$. Use $E = k \times k$ and $z = (y, bx + cy)$, so $N(z) = q(x, y)$. Then $\beta = (e - cd/b, d/b)$ gives the same linear term as $\mathrm{Tr}(\beta z)$ and completing the norm gives $N(z + \tau(\beta)) + r - N(\beta)$. In both cases the allowable z still come from the original two input coordinates. The norm identity alone permits neither independent choices of the two coordinates of a product algebra nor arbitrary unequal shifts of a Cartesian square [11, §§2.1–2.2].

C.2 Mixed-signature fields, return cutoffs and candidate units

Suppress j, write $d = [F : \mathbb{Q}]$, and suppose $\mathrm{rd}(F) \leq D_0$. The cubic $s^3 + s + 1 = 0$ is irreducible over $\mathbb{Q}$ by the rational-root test and has only one real root since its derivative is positive. A subfield of a totally real field is totally real, whereas $\mathbb{Q}(s)$ has two nonreal embeddings. If the cubic were reducible over $F \subset \mathbb{R}$, it would have a root there, necessarily its real root s. It therefore remains irreducible over F. Each embedding of F has one real extension and one complex conjugate pair in $L = F(s)$. The chosen real copy of L does not contain ω, so $E = L(\omega)$ doubles the degree. In the convention counting one place per conjugate pair, the signatures are (d, d) and $(0, 3d)$.

The relative discriminants divide those of the displayed orders generated by s and ω. Their defining polynomials have discriminants -31 and -3, respectively. Hence $\mathrm{rd}(L) \leq D_* = 31^{1/3} D_0$ and $\mathrm{rd}(E) \leq D_E = \sqrt{3} D_*$. For fixed $T \geq T_0 = 2\sqrt{D_*}$ the box in Section 4.2 has the source bounds

$$\left(\frac{(\pi/2)T^3}{D_*^{3/2}}\right)^d \leq |C_j(T)| \leq (2T + 1)^{3d}.$$

The lower bound comes from one real interval and one complex disc per base degree in a translated half-window, followed by subtraction of one selected point. Integer-norm separation gives the upper bound. At $T \geq T_0$ the lower base exceeds one [12, §2 and mixed-signature prerequisites].

Every unit returning a nonzero input lies within the cutoff. Fix a nonzero pair $\mathbf{p} = (x, y)$ and suppose both $\mathbf{p}$ and $M_u\mathbf{p}$ satisfy the window bound T for an order unit of relative norm one. Set $w = Q_\omega(x, y)$. For each complex place σ_i of L, the two extensions evaluate $x - \omega y$ at conjugate roots of unity, and their product is $\sigma_i(w)$. The two multipliers on these coordinates have moduli e^{t_i} and e^{-t_i}, where $t_i = \log |v_i^+(u)|$. At source and target each coordinate has modulus at most $2T$. The source coordinate with the larger multiplier must therefore have modulus at most $2Te^{-|t_i|}$. Thus $|\sigma_i(w)| \leq 4T^2 e^{-|t_i|}$.

At every real embedding $0 < \sigma(w) \le 3T^2$. Positivity follows from positive definiteness and injectivity of the embedding on a nonzero pair. The absolute norm of the nonzero algebraic integer w has absolute value at least one. Complex places occur twice in that norm, giving

$$\sum_{i=1}^{d} \log|\sigma_i(w)| \ge -\tfrac{d}{2}\log(3T^2).$$

Consequently

$$\sum_i |t_i| \le d\log(4T^2) - \sum_i \log|\sigma_i(w)| \le \tfrac{d}{2}\log(48T^6).$$

This proves the cutoff for every nonzero return. The origin, fixed by the entire group, was separated in (18) [12, mixed-signature field companion, MS-F6, (20)–(23)].

An exponential lower bound for the candidate-unit count. The population theorem uses a second calculation. Its explicit constants are

$$B_0 = 2\sqrt{D_E}, \quad g = \log B_0, \quad J_0 = 64B_0^{12}(2B_0^2+1)^3,$$
$$K_E = 6\mathrm{e}gJ_0, \quad K_{\mathcal{R}} = 27K_E, \qquad T_* = \max\{T_0, \exp(2K_{\mathcal{R}}/3)\}.$$

Here e denotes the base of natural logarithms. The proof uses boxes in E with reciprocal complex radii. Minkowski's theorem gives elements approximating directions in the relative logarithmic space. Tracking their principal ideals and their exact relative norms identifies quotients that are actual norm-one units. Covering this space modulo the log lattice gives a maximal-order covolume bound K_E^d. The bound 3^{3d} for the index cost of passing to order units gives the covolume bound $K_{\mathcal{R}}^d$. The proof applies to the indicated mixed signature [12, §3 and arithmetic companion].

If w_j is the order of the torsion kernel of the logarithm on these units, centered lattice counting gives

$$|U_j| \ge \frac{w_j(dH_{\mathrm{cut}}(T_*))^d}{d!K_{\mathcal{R}}^d} \ge w_j\left(\frac{H_{\mathrm{cut}}(T_*)}{K_{\mathcal{R}}}\right)^d \ge 2^d.$$

The last comparison uses $H_{\mathrm{cut}}(T_*) \ge 3\log T_* \ge 2K_{\mathcal{R}}$. This calculation counts candidates. The remaining orbit estimate in [13] would require fixed $C, \gamma > 0$ with $o_j \le C(|C_j|^2 - 1)e^{-\gamma d_j}$ at all sufficiently large levels. The proved size bound for U_j supplies no such estimate on occupied orbits.

C.3 An image-size lower bound for the balanced-factor sets

Fix a depth $r \ge 0$ and put $q = 7^r$, $\xi = e^{2\pi i/q}$, $\omega = e^{2\pi i/3}$ and $\Delta_\omega = 1 + 2\omega$. For $1 \le h \le 8$ and digits $i_\ell \in \{-3,-2,-1,0,1,2,3\}$ define

$$c_{h,\mathbf{i}} = h\prod_{\ell=0}^{r-1}(2 - \xi^{i_\ell 7^\ell}), \qquad B_r = \{c_{h,\mathbf{i}}\xi^a : a \in \mathbb{Z}/q\mathbb{Z},\ h, \mathbf{i} \text{ as above}\}.$$

Multiplication by ξ^a rotates a complex number by $2\pi a/q$. The product is 1 for $r = 0$. With bars denoting complex conjugation, the remaining blocks and the real input are

$$D_r = \{\bar{b} - b : b \in B_r\}, \qquad V_r = B_r \cup (-B_r) \cup D_r,$$
$$A_r = \{X(v) : v \in V_r\}, \quad X(v) = \frac{\omega\bar{v} - \omega^2 v}{\Delta_\omega}.$$

Conjugation negates both numerator and denominator in X, so these scalars are real. The source proves the three blocks disjoint, X injective on $\mathbb{Q}(\xi)$, $|B_r| = 8q^2$, and $16q^2+1 \le n = |A_r| \le 24q^2-7$ [15, §2 and algebra companion]. The inputs are actual scalars with duplications removed.

To count complete fibers, define $m(v,w) = 2vw + v\bar{w} + \bar{v}w - \bar{v}\bar{w}$ and $k(v,w) = m(v,v) + m(v,w) + m(w,w)$. The identity

$$X(k(v,w)) = 3Q_\omega(X(v), X(w))$$

allows the source to test output equality against all targets in V_r^2. Its tower and elimination arguments prove fixed integers $M \ge 1, r_*$ such that, for every $r \ge r_*$, at least $32q^4$ ordered sources in B_r^2 have at most M targets with the same value in the entire V_r^2. Counting these fibers gives

$$|Q_\omega(A_r, A_r)| \ge 32q^4/M \ge n^2/(18M).$$

The constants are proved to exist. No numerical value of M is needed to exclude fixed power saving on this family.

C.4 Quartic local-density stabilization and global counting limits

Lift $\alpha \in \{3,5,6\}$ to the indicated integer and put $\mathcal{B} = \alpha + \mathfrak{p}$ in the complete valuation ring of Section 4.3. The change $(x,y) \mapsto (v,x)$ with $v = x+2y$ maps $\mathcal{B}^2$ bijectively onto $(3\alpha+\mathfrak{p}) \times (\alpha+\mathfrak{p})$, since $y = (v-x)/2$. Define $g(v) = v^4 - v/2$. Then $P_4(x,y) = g(v) + (7/2)x$ and independence gives the exact set

$$P_4(\mathcal{B}, \mathcal{B}) = g(3\alpha + \mathfrak{p}) + (7/2)\alpha + \mathfrak{p}^4.$$

Indeed $(7/2)\mathfrak{p} = \mathfrak{p}^4$, so the last term fills every digit beyond the first four.

Let $v_0 = 3\alpha$. Its cube equals 1 modulo 7, so $g'(v_0) = 4v_0^3 - 1/2$ belongs to $\mathfrak{p}^3$. For $u \in \mathfrak{p}$, Taylor expansion modulo $\mathfrak{p}^4$ leaves $Au^2 + Bu^3$ with $A = 6v_0^2$ a unit and $B = 4v_0$. Write $u = \pi b + \pi^2 c \bmod \mathfrak{p}^3$ for a uniformizer π and digits $b, c \in \{0, \dots, 6\}$. The remaining terms are

$$Ab^2\pi^2 + (2Abc + Bb^3)\pi^3 \pmod{\mathfrak{p}^4}.$$

When $b = 0$ there is one class. For $b \ne 0$ the leading digit has three possible square multiples, and c fills all seven next digits for each. This gives $1 + 3 \cdot 7 = 22$ classes modulo $\mathfrak{p}^4$ and four classes modulo $\mathfrak{p}^3$. Every higher tail is filled, proving $N_1 = 4$ and $N_m = 22 \cdot 7^{3m-4}$ for $m \ge 2$ [16, §§2–4].

The source compares these counts with an actual input in its own totally real tower. With $d = [F_j : \mathbb{Q}]$ and fixed root discriminant $D \ge 7^{2/3}$, each of the $d/3$ completions above 7 is the source cubic. Write

$$7O_{F_j} = \prod_{k=1}^{d/3} \mathfrak{P}_{j,k}^3, \quad \mathfrak{I}_j = \prod_{k=1}^{d/3} \mathfrak{P}_{j,k}, \qquad X = 2\,7^{1/3}\sqrt{D}.$$

The input is $C_j = -1 + \{u \in \mathfrak{I}_j : |\sigma(u)| \le X \text{ for every real } \sigma\}$. In particular its local residues lie over $-1 \equiv 6 \pmod 7$. The ideal-lattice input bound is $|C_j| \ge (X/(7^{1/3}\sqrt{D}))^d = 2^d$. On I^2, where $I = [-1-X, -1+X]$, the source gives $W = 81(X+1)^4 + 3X/2 + 27/16$. CRT bounds the possible output classes by $N_m^{d/3}$. Within one class integer norms separate distinct outputs in the maximum norm by at least 7^m. Thus the full-interval packing bound is $N_m^{d/3}(1 + W/7^m)^d$. Comparing it with the squared input lower bound yields the rate in Section 4.3 [16, §5, (5.1)–(5.5)].

In the separate hybrid proposal, inputs gt, hu use units g, h and additive factors t, u. For $g \ne 0$,

$$P_4(gt, hu) = g^4\big(t + 2(h/g)u\big)^4 + g\big(3t - (h/g)u\big).$$

The same ratio and the same unit occur in both terms, with different powers. Separate estimates for products and sums do not control the number of values of this correlated expression. A bound for the complete union over all realized ratios and both additive inputs is still missing [9].

C.5 Cubic image lower bounds and the rich-fiber example

Fix the witness from Section 4.4 with t distinct completely split primes, all $q \equiv 1 \pmod{12}$. Write $N = |B_{F_j}(Q/8)|$, $d = d_j$ and set

$$\rho = \prod_{q|Q} \frac{q+1}{2q}, \qquad \Theta_{P_2} = D\rho(164 + 64/Q), \qquad \delta = \tfrac{1}{2}\log Q - t\log 4 - \log(32\sqrt{D}).$$

The numerical condition explaining the failure of transfer is $\Theta_{P_2} < 1$. It is the condition of the specific quadratic estimate $|P_2(A_j, A_j)| \le \Theta_{P_2}^d N^2$. In this prime regime the source proves that it forces $\delta > 0$ [17, complementary-prime checkpoint, §F]. Indeed $\rho > 2^{-t}$ and $D \ge 1$ give $t \ge 8$ and $D < 2^t/164$. The kth prime selected in increasing order exceeds $12k$, so $Q > 12^t t!$. Substitution gives $e^{-2\delta} < (256/41)(8/3)^t/t!$, which is below one at $t = 8$ and decreases thereafter.

A collision between different algebraic differences u, v requires $u^3 \equiv v^3 \pmod{Q^2 \mathcal{O}_{F_j}}$, with $u, v \in B_{F_j}(Q/4)$. The key arithmetic estimate in [17, complementary-prime checkpoint, §D, (D.1)] bounds these ordered difference pairs by $(4^{t+1}\sqrt{Q})^d$, including singular residue classes. Once the first input pair and both differences are fixed, equality of the cleared cubic outputs determines the second sum and hence the second pair. For each fixed first difference there are at most N choices for the first pair. Equal differences contribute just the N^2 identical-pair collisions. The resulting collision energy is therefore at most

$$N^2 + N(4^{t+1}\sqrt{Q})^d \le N^2(1 + e^{-\delta d}),$$

where the last inequality uses $N \ge (Q/(8\sqrt{D}))^d$. Cauchy–Schwarz now gives the actual-image lower bound in Section 4.4.

For the separate rich-fiber construction, write $f_\pm(i) = 36i^3 \pm 3i + 2$. The shared value -12 follows from $f_+(i) - f_-(i) = 6i$ and $f_+(i) + f_-(i) = 72i^3 + 4$. The checkpoint's finite-avoidance argument can be seen directly on adjoining two new vertices. For an old vertex a the polynomial $P_-(f_\epsilon(i), a)$ has coefficients of i^9, i^7, i^6 equal to $46656, 11664\epsilon, 7776 - 3888a$ for $\epsilon \in \{-1, 1\}$. They distinguish the old vertex, sign and orientation (the reversed orientation negates these coefficients). Among the four new/new outputs the two diagonals are $-216i^3 \mp 18i - 12$, the reversed off-diagonal is $-432i^3 - 12$, and the designated off-diagonal is -12. Thus every unwanted equality with an old output or another new output is a nonzero polynomial equation. Excluding its finitely many roots and taking both new vertices above the previous ones proves the multiplicity count by induction. The original construction selects indices in an infinite arithmetic progression imposing further congruences for its number fields. Finitely many exclusions preserve a choice there [17, rich-fiber checkpoint, §§B.1–B.2 and C].

C.6 Exact radical-parameter selection and the open growth condition

Exact one-step parameter selection. For a finite symmetric input A and legal $C > M = \max_{a\in A} |a|$, define the signed output set

$$\mathcal{T}_C(A) = \left\{2C + a + b \ \pm \sqrt{(C + a)(C + b)} : a, b \in A\right\}.$$

Each unordered pair of old inputs, allowing equal entries, has two branch labels corresponding to the two signs. A direct substitution gives

$$Q_\omega(\mathrm{Rad}_C(A), \mathrm{Rad}_C(A)) = (C - M)^{-1}\mathcal{T}_C(A).$$

The two branches of the same unordered pair differ because the root is positive. Before any coincidences between distinct pairs there are $|A|(|A| + 1)$ labels. A legal parameter is helpful if two of these distinct labels coincide.

The following finite selection rule uses exact real algebraic arithmetic on the actual sequence starting at $\{-1, 1\}$ [14, §§3, 4.2, (CP)].

1. Enumerate distinct unordered old pairs with repetitions allowed. For two different pairs let (s, p) and (t, q) be their sums and products, and put $D = s - t$, $E = p - q$. Skip $D = 0$, since then equality of signed root terms would force the products equal as well, giving the same unordered pair.

2. For $D \neq 0$, enumerate the real roots $C > M$ of

$$\begin{aligned} 0 = \ & -3D^2C^2 + [2DE - 2D^2(s + t)]C \\ & + E^2 - 2D^2(p + q) + D^4. \end{aligned}$$

   This quadratic follows by squaring the equality of the two signed branches twice. Check the two original signs on each side before retaining a collision. There are finitely many unordered pairs and the leading coefficient is nonzero.

3. Combine all retained roots with $M + 1$ and remove duplicates. For each listed C count the distinct values in $\mathcal{T}_C(A)$ exactly. Choose the least listed C attaining the smallest count and form $\mathrm{Rad}_C(A)$.

Every unlisted legal choice has the full number of distinct labels. This proves the one-step comparison used in the adaptive rule. The recorded first choice $C = 5/3$ provides a direct check of the normalization. From $A_0 = \{-1, 1\}$ the two positive roots are 1 and 2, hence $A_1 = \{-2, -1, 1, 2\}$. Optimizing this single successor says nothing about the sizes of the images of its later descendants.

An unresolved sufficient condition for power saving. One explicit sufficient bound, still open, concerns the actual levels A_k. Write $n = n_k$, $s = s_k$ and $F_k^{\mathrm{rad}} = \mathbb{Q}(A_k)$. The premise asks for fixed H, k_0 such that at every $k \geq k_0$ one can choose a legal $C \in F_k^{\mathrm{rad}}$ and a subset $B \subseteq A_k$ omitting at most $H\sqrt{n}$ indices. It also requires positive $\gamma, u_a \in F_k^{\mathrm{rad}}$ with $C + a = \gamma u_a^2$ for every $a \in B$ and, writing $U = \{\pm u_a : a \in B\}$,

$$\left|\gamma Q_\omega(U, U)\right| \leq \tfrac{15}{4} s + H n^{3/2}.$$

If $e = n - |B|$, at most $n(n+1) - (n-e)(n-e+1) = e(2n-e+1)$ branch labels involve an omitted index. For the others, a common scaling by $(C - M)^{-1}$ preserves cardinality. The one-step minimum would therefore satisfy $s_{k+1} \leq (15/4) s_k + 4H n_k^{3/2}$. Since $n_k = 2^{k+1}$ and $2^{3/2} < 15/4$, summing the resulting geometric series gives $s_k \leq C_1 (15/4)^k$ for a fixed C_1. This proves the implication for every $0 < \varepsilon < \log_2(16/15)$. The uniform choice and signed-image premise are unresolved [14, §7, (RC15/4)].

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